

SUMMARY

Licensed Professional Engineer and PMP-certified technical leader with 15+ years of progressively responsible federal service delivering complex engineering, project management, and risk analysis solutions across the Department of Energy (DOE), Department of Defense (DOD), and Department of Homeland Security (DHS). Recognized expert in nuclear, electrical, and infrastructure systems supporting national security, environmental remediation, and mission-critical facilities, with oversight of projects and portfolios exceeding \$8 billion. Demonstrated success advising senior executives on risk-informed decision-making, lifecycle cost analysis, regulatory compliance, and emerging technology integration, inducing responsible AI governance. Recipient of multiple awards for technical leadership, innovation, and sustained mission impact.

PROFESSIONAL EXPERIENCE

DOE - ENVIRONMENTAL MANAGEMENT, OFFICE OF PROJECT MANAGEMENT (EM-5.22)

Washington, DC

General Engineer (GS-14)

September 2023 - Present

- Serve as senior technical advisor supporting DOE's \$8B+ Cold War nuclear cleanup mission, addressing highly complex environmental, safety, cost, and lifecycle challenges originating from the Manhattan Project legacy.
- Conduct independent technical, cost, and risk reviews of Federal Site Lifecycle Estimates (FSLEs), ensuring alignment with DOE Order 413.3B, GAO cost-estimating standards, and risk-informed decision frameworks.
- Perform cross-site analyses and benchmarking across major DOE cleanup sites (Hanford, Los Alamos, Savannah River, Oak Ridge, and Idaho National Laboratory), identifying cost drivers, schedule risks, and technology gaps, and opportunities for performance improvement.
- Evaluate engineering assumptions, contingency methodologies, escalation models, and risk registers to manage senior-level funding, sequencing, and portfolio tradeoff decisions.
- Lead end-to-end execution of mission-critical projects using Agile and Lean Six Sigma, while advising senior leadership on complex technical and strategic decisions.
- Drive data-informed improvements by analyzing technical and operational challenges, developing dashboards and visualizations, and leading change initiatives to optimize performance and efficiency.
- Serve on DOE AI Governance Team, DOE AI Hub, and GSA AI Community of Practice, contributing to development of policy, controls, and implementation guidance for responsible AI use in federal program management, analytics, and decision support.
- Present senior DOE leadership and cross-functional stakeholders on findings, risks, and recommendations affecting long-term cleanup strategy and federal budget formulation.
- Awarded **2024 & 2025 DOE EM individual Bonus Award** for technical leadership and mission impact.

DOD - NAVAL SEA SYSTEMS COMMAND, CARRIER ENGINEERING TEAM (SEA 05V2)

Washington, DC

General Engineer (NH-04)

July 2023 - September 2023

- Provided systems engineering and technical risk oversight across electrical, digital network, and cybersecurity domains for nuclear-powered aircraft carriers (CNV-68 through CNV-77).
- Advised leadership on certification, testing, and risk management considerations impacting mission assurance and lifecycle sustainment.

DOD - WASHINGTON HEADQUARTERS SERVICES, ENGINEERING & ARCHITECTURE DIVISION (EAD)

Washington, DC

General Engineer (GS-14)

May 2023 - July 2023

- Provided technical oversight for sustainment and modernization projects, facilitating the Pentagon and Mark Center.
- Managed multidisciplinary engineering coordination, lifecycle analyses, and executive-level project risk briefings.
- Awarded **2023 WHS Command Coin** and **Quality Step Increase**.

DOD - WASHINGTON HEADQUARTERS SERVICES, STANDARD & COMPLIANCE DIVISION (SCD)

Washington, DC

Electrical Engineer (GS-13)

September 2020 - May 2023

- Led design reviews and modernization of electrical and life safety systems for the Pentagon (6.5M sq. ft.) and Mark Center (1.8M sq. ft.), managing facility portfolios valued at \$300M+.
- Chaired Pentagon Electrical Subcode and Controls/Cybersecurity Subcode Committees, establishing technical standards and compliance requirements across DOD enterprise facilities.
- Led energy resilience and sustainability initiatives, including EV charging infrastructure, solar integration, and cyber-secure building control systems, aligning projects with Executive Orders and congressional mandates.
- Provided technical guidance to senior leadership on modernization strategies, risk mitigation, and long-term infrastructure planning.
- Awarded **2022 WHS Director's Award Nomination**, **SCD Command Coin**, and **Quality Step Increase**.

DHS - U.S. COAST GUARD MARINE SAFETY CENTER, ELECTRICAL BRANCH (E-2)

Washington, DC

Electrical Engineer (GS-13)

February 2016 - September 2020

- Served as subject matter expert in marine electrical systems, conducting plan reviews and approving designs for commercial vessels operating under U.S. and international regulations.
- Specialized in lithium-ion battery systems, fuel cells, LNG dual-fuel propulsion, and hybrid/electric vessels, supporting first-of-their-kind approvals in the United States.
- Reviewed electrical systems for SpaceX rocket landing platform vessels and nation's first all-electric and hybrid commercial vessels.
- Awarded **2019 Civilian Service Commendation Medal** and multiple **Individual Bonus Awards**.

DHS - U.S. COAST GUARD YARD, ELECTRICAL & ELECTRONICS BRANCH (E/E)

Baltimore, MD

Electrical Engineer (GS-12)

September 2010 - February 2016

- Designed, modified, and modernized electrical and electronic systems for Coast Guard Cutters.
- Led multidisciplinary engineering teams supporting \$14M+ Service Life Extension Programs (SLEPs).
- Developed engineering calculations, technical drawings, specifications, and procurement documentation.
- Awarded **2015 Silver Award for Outstanding Professional (Technical/Scientific)** and multiple **Individual Bonus Awards**.

EDUCATION**Johns Hopkins University - Whiting School of Engineering**

Baltimore, MD

Master of Science in Artificial Intelligence (M.S.) - GPA - Candidate

University of Maryland - Robert H. Smith School of Business

College Park, MD

Master of Business Administration (M.B.A), Finance Focus - GPA 3.84

University of Maryland - A. James Clark School of Engineering

College Park, MD

Master of Engineering in Electrical & Computer Engineering (M.Eng.)

Communications & Signal Processing Focus - GPA 3.87

University of Missouri - St. Louis / Washington University in St. Louis Joint Engineering

St. Louis, MO

Bachelor of Science in Electrical Engineering (B.S.), Mathematics Minor, Cum Laude - GPA 3.45

CERTIFICATIONS & LICENSES

- Professional Engineer (P.E.) - District of Columbia
- Project Management Professional (PMP) - PMI
- Certified Energy Manager (CEM) - AEE
- Certified Facility Manager (CFM) - IFMA
- Leadership in Energy and Environmental Design Green Associate (LEED GA) - GBCI
- Lean Six Sigma Black Belt (LSS-BB) - CSSC
- Federal Project Director Level I (FPD I) - DOE
- Federal Acquisition Certification for Contracting Officer Representative Level III (FAC-COR III)
- Federal Acquisition Certification for Program/Project Managers Level III (FAC-P/PM III)
- Construction Quality Management for Contractors (CQM-C)
- Secret Clearance (Inactive, DOD, 2021)

TECHNICAL SKILLS

- Engineering & Analysis: AutoCAD, Revit, SKM, ETAP, AGI32, Visual Lighting, MATLAB, PSPIICE, VHDL
- Data & Programming: Python, R, SQL, C++
- Cloud & DevOps: AWS, Azure, Google Cloud Platform, Docker
- Project & Visualization: Microsoft Power Platform, Adobe Creative Suite, Tableau
- Languages: English (Advanced) Chinese (Advanced) Japanese (Novice)

PROFESSIONAL AFFILIATIONS

- Engineering & Technical Societies: IEEE, AME, ASCE, NFPA, ANS, NSPE
- Project & Facilities Management: PMI, IFMA
- Energy & Sustainability: AEE, USGBC